INCH-POUND

A–A–59590/8A
17 August 2005
SUPERSEDING
A-A-59590/8
24 September 2001

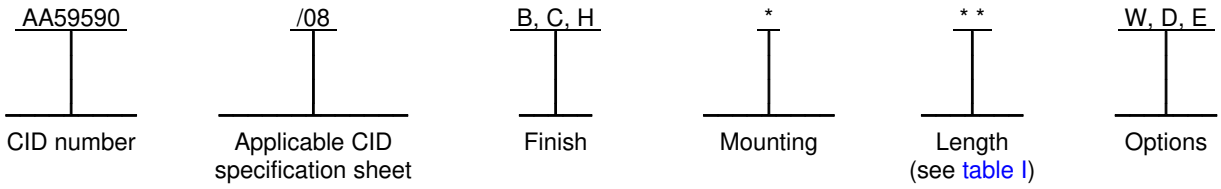
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE, WITH VISUAL LOCK INDICATION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59590/08CM68W is the PIN for a chemical film finished, 6.8 inch (180.3 mm) long card holder with visual lock indication. The card holder also features three tapped mounting holes for use with M2.5 x.45 metric fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G–rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces. Card holders shall have a visual indicator to show when the assembly is in its relaxed (unlocked) state.

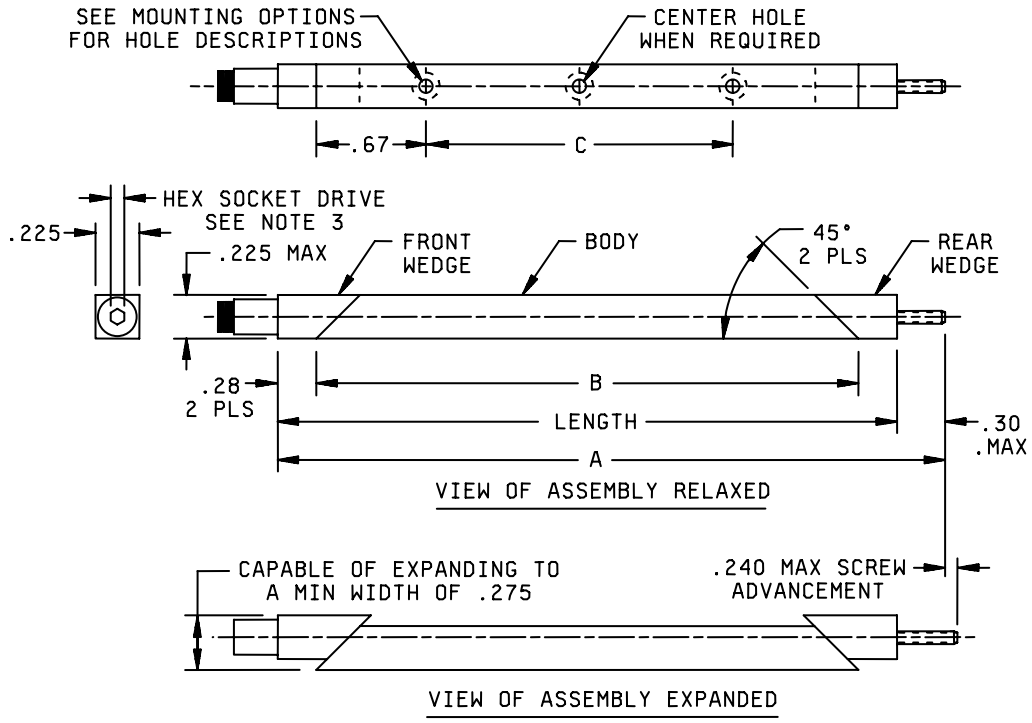
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table 1, and A–A–59590.

Visual lock indicator. The visual lock indicator, intended to show when the card holder is in its relaxed unlocked state, shall be as specified in figure 2. When the card holder is in the unlocked (relaxed) state, the end of the actuating screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.070	1.78	.220	5.59	.225	5.72	.28	7.1
.20	5.1			.275	6.99	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.51 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.

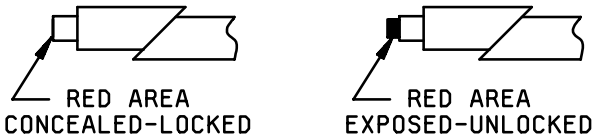


FIGURE 2. Visual lock indicator.

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Finish. The finish options “B” (black anodize), “C” (chemical film), or “H” (hard black anodize) shall be as specified in A–A–59590.

Mounting. The mounting options shall be as specified in A–A–59590. The mounting options available for this specification sheet are as follows: “J” or “K” (rivet mounting holes), “N” (no mounting holes), “T” or “R” (tapped 0-80 UNF 2B holes), “V” or “S” (tapped 2-56 UNC 2B holes), or “G” or “M” (tapped M2.5 x.45 metric holes). Card holders using mounting option “J” or “K” may be shipped unassembled.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to mm) deep.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension “LENGTH” ± .02 (0.5)	Dimension “A” (maximum)	Dimension “B” ± .03 (0.8)	Dimension “C” ± .005 (0.13)
18	1.8 (45.7)	2.10 (53.3)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.60 (66.0)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.10 (78.7)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.30 (83.8)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.60 (91.4)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.80 (96.5)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.10 (104.1)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.60 (116.8)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.10 (129.5)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.60 (142.2)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.10 (154.9)	5.24 (133.1)	3.90 (99.1)
63	6.3 (160.0)	6.60 (167.6)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.10 (180.3)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.60 (193.0)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.10 (205.7)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.60 (218.4)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.10 (231.1)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

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Options. Card holders can have the following options. Card holders not requiring the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 3). The use of a lockwasher and flat washer located under the screw head will provide additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

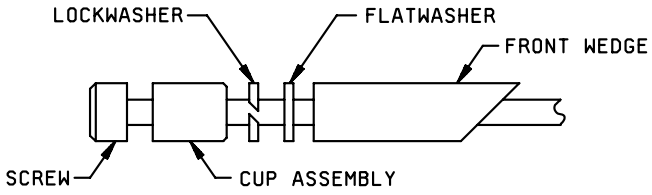


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

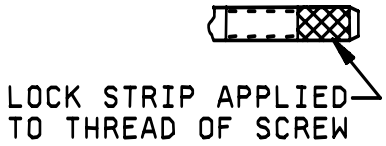


FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	APW Electronic Solutions 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: sales.electronicsolutions@apw.com Uniform Resource Locator (URL): www.apw.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers as shown in table II. The CID PINs listed in table II are only for length designator "28". See table III for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	42VI–6	CJ28	V230–2.80H
CN28	42VI–6–A	CN28	V230–2.80
CT28	42VI–6–T	CT28	V230–2.80T0
CV28	42VI–6–S	CV28	V230–2.80T2
CG28	42VI–6–M	CG28	MV230–2.80TM2.5
CJ28E	42VI–6–L	CJ28E	V230–2.80HL
CN28E	42VI–6–A–L	CN28E	V230–2.80L
CT28E	42VI–6–T–L	CT28E	V230–2.80T0L
CV28E	42VI–6–S–L	CV28E	V230–2.80T2L
CG28E	42VI–6–M–L	CG28E	MV230–2.80TM2.5L
CJ28W	42VI–6–LF	CJ28W	WV230–2.80H
CN28W	42VI–6–LF–A	CN28W	WV230–2.80
CT28W	42VI–6–LF–T	CT28W	WV230–2.80T0
CV28W	42VI–6–LF–S	CV28W	WV230–2.80T2
CG28W	42VI–6–LF–M	CG28W	MWV230–2.80TM2.5

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 52094
CJ28D	42VI–6–LF–L	CJ28D	WV230–2.80HL
CN28D	42VI–6–LF–A–L	CN28D	WV230–2.80L
CT28D	42VI–6–LF–T–L	CT28D	WV230–2.80T0L
CV28D	42VI–6–LF–S–L	CV28D	WV230–2.80T2L
CG28D	42VI–6–LF–M–L	CG28D	MWV230–2.80TM2.5L
BJ28	42VI–6–B	BJ28	VA230–2.80H
BN28	42VI–6–B–A	BN28	VA230–2.80
BT28	42VI–6–B–T	BT28	VA230–2.80T0
BV28	42VI–6–B–S	BV28	VA230–2.80T2
BG28	42VI–6–B–M	BG28	MVA230–2.80TM2.5
BJ28E	42VI–6–B–L	BJ28E	VA230–2.80HL
BN28E	42VI–6–B–A–L	BN28E	VA230–2.80L
BT28E	42VI–6–B–T–L	BT28E	VA230–2.80T0L
BV28E	42VI–6–B–S–L	BV28E	VA230–2.80T2L
BG28E	42VI–6–B–M–L	BG28E	MVA230–2.80TM2.5L
BJ28W	42VI–6–B–LF	BJ28W	WVA230–2.80H
BN28W	42VI–6–B–LF–A	BN28W	WVA230–2.80
BT28W	42VI–6–B–LF–T	BT28W	WVA230–2.80T0
BV28W	42VI–6–B–LF–S	BV28W	WVA230–2.80T2
BG28W	42VI–6–B–LF–M	BG28W	MWVA230–2.80TM2.5
BJ28D	42VI–6–B–LF–L	BJ28D	WVA230–2.80HL
BN28D	42VI–6–B–LF–A–L	BN28D	WVA230–2.80L
BT28D	42VI–6–B–LF–T–L	BT28D	WVA230–2.80T0L
BV28D	42VI–6–B–LF–S–L	BV28D	WVA230–2.80T2L
BG28D	42VI–6–B–LF–M–L	BG28D	MWVA230–2.80TM2.5L
HJ28	42VI–6–B3	HJ28	VHA230–2.80H
HN28	42VI–6–B3–A	HN28	VHA230–2.80
HT28	42VI–6–B3–T	HT28	VHA230–2.80T0
HV28	42VI–6–B3–S	HV28	VHA230–2.80T2
HG28	42VI–6–B3–M	HG28	MVHA230–2.80TM2.5

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 52094
HJ28E	42VI–6–B3–L	HJ28E	VHA230–2.80HL
HN28E	42VI–6–B3–A–L	HN28E	VHA230–2.80L
HT28E	42VI–6–B3–T–L	HT28E	VHA230–2.80T0L
HV28E	42VI–6–B3–S–L	HV28E	VHA230–2.80T2L
HG28E	42VI–6–B3–M–L	HG28E	MVHA230–2.80TM2.5L
HJ28W	42VI–6–B3–LF	HJ28W	WVHA230–2.80H
HN28W	42VI–6–B3–LF–A	HN28W	WVHA230–2.80
HT28W	42VI–6–B3–LF–T	HT28W	WVHA230–2.80T0
HV28W	42VI–6–B3–LF–S	HV28W	WVHA230–2.80T2
HG28W	42VI–6–B3–LF–M	HG28W	MWVHA230–2.80TM2.5
HJ28D	42VI–6–B3–LF–L	HJ28D	WVHA230–2.80HL
HN28D	42VI–6–B3–LF–AL	HN28D	WVHA230–2.80L
HT28D	42VI–6–B3–LF–TL	HT28D	WVHA230–2.80T0L
HV28D	42VI–6–B3–LF–SL	HV28D	WVHA230–2.80T2L
HG28D	42VI–6–B3–LF–ML	HG28D	MWVHA230–2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Example of PIN with available length designators.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/08	Vendor similar designator or type part number 1/ 2/ CAGE 52094
CG18W	42VI–4–LF–M	CG18W	MV230–1.80TM2.5
CG23W	42VI–5–LF–M	CG23W	MV230–2.30TM2.5
CG28W	42VI–6–LF–M	CG28W	MV230–2.80TM2.5
CG30W	3/	CG30W	MV230–3.00TM2.5
CG33W	42VI–7–LF–M	CG33W	MV230–3.30TM2.5
CG35W	3/	CG35W	MV230–3.50TM2.5
CG38W	42VI–8–LF–M	CG38W	MV230–3.80TM2.5

See footnotes at end of table.

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TABLE III. Example of PIN with available length designators – Continued.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/08	Vendor similar designator or type part number 1/ 2/ CAGE 52094
CG43W	42VI–9–LF–M	CG43W	MV230–4.30TM2.5
CG48W	42VI–10–LF–M	CG48W	MV230–4.80TM2.5
CM53W	42VI–11–LF–M	CM53W	MV230–5.30ETM2.5
CM58W	42VI–12–LF–M	CM58W	MV230–5.80ETM2.5
CM63W	42VI–13–LF–M	CM63W	MV230–6.30ETM2.5
CM68W	42VI–14–LF–M	CM68W	MV230–6.80ETM2.5
CM73W	42VI–15–LF–M	CM73W	MV230–7.30ETM2.5
CM78W	42VI–16–LF–M	CM78W	MV230–7.80ETM2.5
CM83W	42VI–17–LF–M	CM83W	MV230–8.30ETM2.5
CM88W	42VI–18–LF–M	CM88W	MV230–8.80ETM2.5

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–007

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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